# Introduction To Lid Disease

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1

## SOAP Format

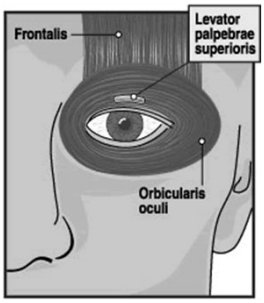
- S = Subjective
  - AKA symptoms
  - Everything the patient says
- O = Objective
  - AKA signs
  - All the doctors observes and all the test results
- A = Assessment
  - AKA diagnosis (or diagnoses)
  - What is wrong with the patient
- P = Plan
  - For each diagnosis, what is going to be done to help the patient

2

## Lid Anatomy

**Muscles**

- Levator
- Mueller's muscle
- Orbicularis

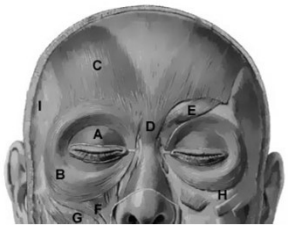



3

## Lid Anatomy

A = orbicularis muscle (palpebral portion)
- Striated
- Eyelid closing muscle - used in blinking and voluntary winking
- Keeps lid tight to globe

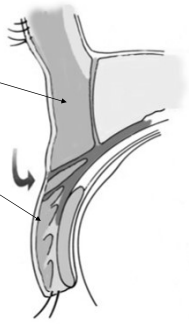
B = orbicularis muscle (orbital portion)
- Striated
- Eyelid closing muscle - used in forced closure




4

## Lid Anatomy

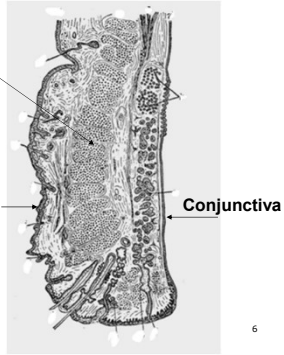
- Orbicularis - orbital portion
- Orbicularis - palpebral portion



5

## Lid Anatomy

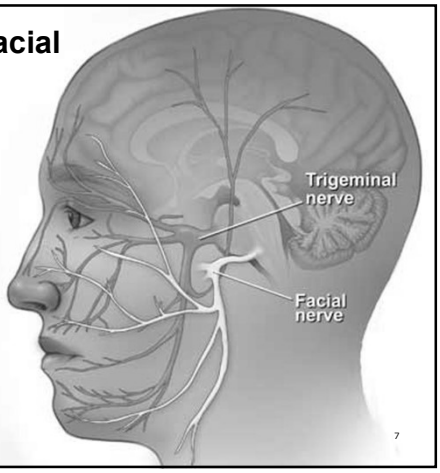
- Orbicularis
  - Innervation is from the facial nerve (CN VII)




6

## CNVII = Facial Nerve

- CNVII causes:
- eyelids to close
- mouth and cheeks to 'smile'
- forehead to 'furrow'




7

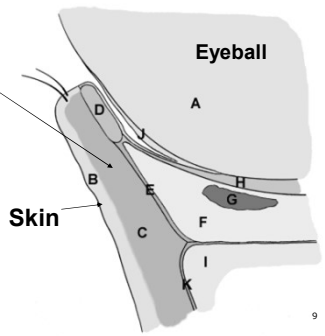
## Bell's Palsy






## Lid Anatomy

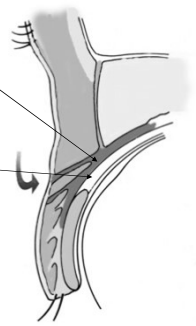
- Orbicularis (lower lid)




9

## Lid Anatomy

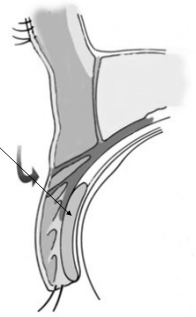
- **Levator aponeurosis**
  - Striated
  - Lid opening muscle
  - Innervated by CN III
  - Fibers pass through the orbicularis to attach to the skin
- **Muller's muscle**
  - Originates from the underside of the levator
  - Smooth
  - Lid opening muscle
  - Innervated by the sympathetic system
  - Inserts into superior tarsus



10

## Lid Anatomy

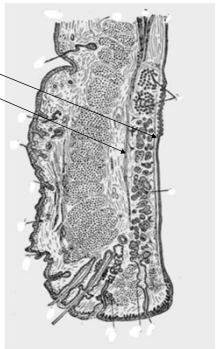
- **Tarsal plate**
  - Dense fibrous connective tissue
  - Responsible for structural integrity of the lids
  - Upper plate about 1 cm high x 2.5 cm wide
  - Lower tarsal plate smaller dimensions



11

## Lid Anatomy

- **Tarsal plate (between arrows)**



12

## Lid Anatomy

- Each tarsus encloses about 25 meibomian glands
- Meibomian gland
  - Sebaceous gland (oil)
  - Provides the oily layer of the tear film
  - Meibomian gland opening just behind the lashes



13

## Lid Anatomy

- Lash follicle and associated sebaceous gland(s) (Zeiss gland)
- Moll gland - sweat



14

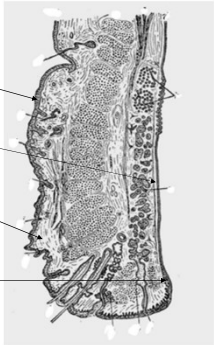
## Lid Anatomy

Skin

Conjunctiva

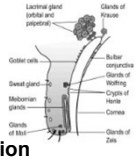
Loose connective tissue

Mucocutaneous junction



15

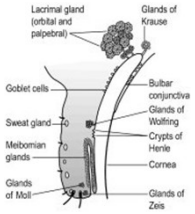
## Eyelid Glands – Conjunctival Side




- Goblet cells
  - In conjunctival epithelium
  - Secrete mucin
  - Abundant at plica semilunaris (caruncle)
  - None on limbus or mucocutaneous junction
- Accessory glands of Krause and Wolfring
  - Krause – near fornix; basal aqueous secretion
  - Wolfring – near tarsus; basal aqueous secretion
  - Both assist lacrimal gland in aqueous secretion
  - Only lacrimal gland has reflex secretion (tearing)

16

## Eyelid Glands – Lid Margin and Skin

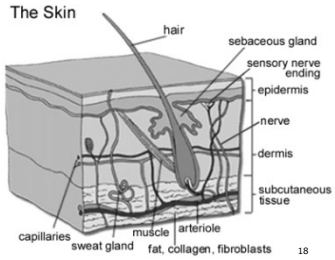



- Meibomian glands
  - Secrete sebum (lipid)
- Gland of Zeiss
  - Unilobular gland
  - Produce oils for the lash follicles
- Gland of Moll
  - aka ciliary gland
  - Anterior to Zeiss glands
  - Produce oils as well
- Sweat gland
  - aka sudoriferous gland

17

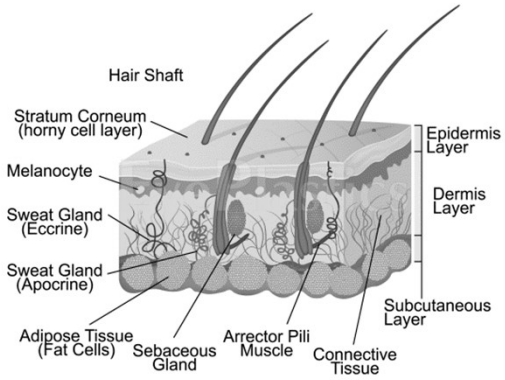
## Lid Anatomy - Skin

- Eyelid skin is the thinnest of the body
  - Epidermis
  - Dermis
  - Hair follicle and associated sebaceous gland
  - Sweat gland
  - Blood supply




18

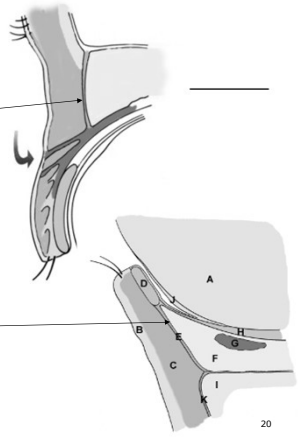
## Skin Anatomy



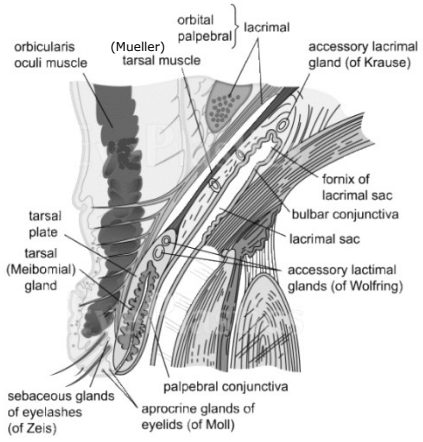

19

## Lid Anatomy

- **Orbital septum**
  - **Attaches peripherally at the periosteum at the orbital margin**
  - **Centrally, it fuses with levator aponeurosis (upper lid) and the tarsal plate (lower lid)**
  - **Acts as diaphragm to separate orbital structures from lid structures**




20




21

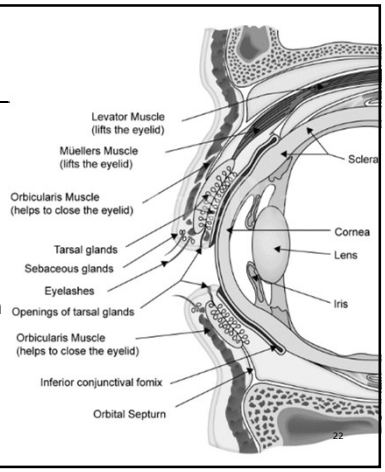
**Eyelids are complex!**

- **Skin**
- **Hair**
- **Fascia**
- **Muscle**
  - **Striated**
  - **Smooth**
- **Glands**
- **Nerves**
- **Conjunctiva**
- **Connective tissue**

**…lots of structures for disease to occur!**

## Lid Diseases

- Position/shape
- Inflammation
- Allergy
- Infection
- Cancer
- Autoimmune
- Gland dysfunction
- Neurological
- Trauma




22

## THANK YOU!

23